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Hybrid renewable system based on solar wind and fuel cell energies coupled with diesel engines for Tunisian climate: TRNSYS simulation and economic assessment

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ABSTRACT

The current investigation examines the feasibility and design of hybrid renewable energy system (HRES) based on wind turbine, photovoltaic, and fuel cell technologies, coupled to diesel engines system to meet the power demand of an off-grid residential community located at the village of Sidi Daoud in Tunisia. Techno-economic study of HRES is carried out to propose the optimized system that satisfies a better performance, enhances the penetration of renewable energy, diminishes the Levelized Cost of Energy (LCOE) and reduces the carbon dioxide gas emission. It is shown that the system guarantees a dumped energy of only 4.7%, less than 0.05% of unmet power and reduced power losses resulting from converting procedure of 1.02% from the total production. From a financial perspective, the proposed model presents a competitive LCOE of \$0.0492/kWh and a renewable fraction of 35.52% with further diminished carbon dioxide emission.

ARTICLE HISTORY

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KEYWORDS

TRNSYS simulation; wind turbine; PV system; fuel cell; diesel generator

1. Introduction

The growth of progressive power demand and rapid diminution of conventional energy sources derived from fossil fuels have raised concerns of future energy supplies (Curtin et al. 2019; Leonard, Michaelides, and Michaelides 2020; Liao et al. 2019; Yue et al. 2019). Additionally, using fossil fuels has resulted in tremendous rises in carbon dioxide (CO₂) emissions with almost 40% of CO₂ increase when comparing with that at the industrial revolution beginning, and which is considered as the crucial cause of global warming (Cormos and Cormos 2017; Owusu and Asumadu-Sarkodie 2016; Ulleberg and Hancke 2020). This atmospheric issue forced the scientists and engineers to search for alternative power sources to generate sustainable energy supply and hence diminish pollution (Abdin and Mérida 2019; Aziz et al. 2015; Duman and Güler 2018). In this vein, the potential for using clean renewable energy technologies for electricity production in Tunisia is important given the abundant wind resources (Ben Amar, Elamouri, and Dhifaoui 2008; Elamouri and Ben Amar 2008) and solar insolation (Trabelsi et al. 2016). Tunisia's wind potential is estimated at 8 GW on an exploitable surface of 1600 km². In 2017, as reported by Tunisian Electricity and Gas Company called in French acronym (STEG) "Société Tunisienne de l'Electricité et du Gaz", the installed capacity in renewable energies represented 3% of the total power, i.e., 311 MW. This installed capacity is divided between wind power (244 MW), hydraulics (62 MW) and photovoltaics (5 MW). Furthermore, Tunisia has a significant energy control potential estimated at almost 100 million toe over the period 2005–2030 from

both the rational use of energy and renewable energies (notably wind and solar). Nevertheless, so far, the use of renewable energy worldwide and especially in Tunisia had been restricted to establishing projects in which single renewable energy source were utilized, further no attention has been paid to large-scale attempts in using renewable power. Moreover, using hybrid technologies for microgrid system, including more than one energy production technology has not been also intensively applied (Elkadeem et al. 2020; Gharibi and Askarzadeh 2019; Saheli, Lari, and Ziarati 2019).

A Hybrid Renewable Energy System (HRES) is an energy system that uses several energy resources containing renewables like solar, wind, geothermal, hydro, ocean, and biomass ... These hybrid renewable systems can be integrated in microgrid power systems that may involve also the renewable, loads for residential, power electrical generators, converters, and control systems. The micro grid power systems may operate as an off-grid (stand-alone system) or grid-tied power system. It is becoming common for stand-alone systems in remote and desert areas thanks to renewable energy technologies advancement and availability of electric converters. Besides, the micro grid power system has numerous benefits for local community goals such as variation of the energy sources, reliability, reduction of the carbon dioxide emissions, economic cost reduction as well as decreasing the impacts of energy outages in communities. Multiple studies dealing with hybrid systems using renewable energies integrated in micro grid power systems can be found in the literature (Johannsen, Østergaard, and Hanlin 2020; Yamashita et al. 2019). For instance, using

TRNSYS simulations, an investigation is carried out by Anoune et al. (2018) to design a PV-Wind-based Hybrid System with the aim of supplying electrical load demands relative to the international university campus located in Rabat town in Morocco. The proposed system allowed providing an optimized hybrid system for two deterministic methods, namely the worst month and a yearly average month methods, by modeling the long-term power performance of multiple configurations tested of the model. The examination of the considered TRNSYS model pinpoints the hybrid system requirements and the cost summary investigation, and, evaluates the environmental and economic impacts.

In the study of Lokar and Virtic (2020), the authors analyzed the yearly energy production of photovoltaic and a pilot battery storage system installed in a typical house situated in Slovenia. It is concluded that complete self-sufficiency can be reached by adding hydrogen fuel cells to the hybrid system. In terms of financial point of view, the authors stated that, by further augmenting the PV performance, such systems involving the hydrogen power are accessible for marketable use despite the higher initial investment, due to the cost elevation of hydrogen storage tank.

Mert, Ekinci, and Demirdelen (2019) studied the influence of partial shading on an off-grid PV/hydrogen system established at ATU University, Adana, in the country of Turkey. The study focused principally on electrolysis in cells with a fluctuating power supply. Analysis of the cell reaction is carried out by monitoring the pressure, voltage, and temperature with further parameters, namely the time variations in oxygen, the hydrogen concentration and the current efficiency. The main outcome of the investigation is to generate hydrogen by means of a part of electrical energy produced from the PV array, and meanwhile to reveal the influence of the electrical energy gained from partial shading of the panels on hydrogen generation. Furthermore, impact of cathode material type is also studied and demonstrates that Cu/NiMo improves hydrogen generation efficiency than other materials considered.

A concentrated wind turbine, solar, and fuel cell power systems have been integrated with an absorption cooling subsystems to supply electrical energy, heating and cooling process, and hydrogen to residential applications and analyzed by Sorgulu and Dincer (2018). The hybrid system components have been involved to control the excess energy through water electrolysis process for production and hydrogen storage. The study results show that the proposed system is efficient, sustainable, and environmentally friendly.

In the paper of Attemene et al. (2020) the authors determined the optimal configuration of a hybrid renewable system integrating a wind turbine considered as the main source with backup energy sources consisting of an alkaline electrolyzer, a PEM fuel cell, and a super-capacitor bank and a battery integrated in an off-grid application. The optimization process has been studied using non-dominated sorted genetic algorithm and based on the system energetic reliability and the total annual cost. The authors also analyzed the total annual cost sensitivity with the wind speed profiles variations and the subsystem component prices. It is seen that with the speed profile variation of wind, a reduction of percentage of up to

65.52% of the initial investment cost is observed which makes the system more gainful in the windy areas. Furthermore, Saleem et al. (2020) applied TRNSYS software to design and optimize a solar-hydrogen co-generation system under warm climate regions and feeble solar insolation to study annual hydrogen produced by the hybrid system. The proposed system mainly includes a PV array system, a fuel cell, an electrolyzer, a battery and a hydrogen storage unit and a controller in the complete loop. Due to lower solar insolation, it is shown that the hydrogen cogeneration system has an efficiency of 7.8% in the cold climate of Fargo North Dakota state. However, an efficiency of 11.8% is provided for hot climate condition of Lahore weather. Using TRNSYS simulation Ghenai, Salameh, and Merabet (2020) designed a renewable hybrid off grid power system to meet the electric loads of 150 houses of residential community located in Sharjah in a desert region. The micro grid hybrid system comprises solar PV panels and a fuel cell system integrated with electrolyzer to produce hydrogen and a DC/AC Inverter. The simulations show that the hybrid power system meets the daily load demand of the residential community. Besides, the authors demonstrated that the technical-economic analysis showed an acceptable penetration of renewable resources with a renewable fraction of 40.2%, an excess percentage of 1.8% from the total power system production, a levelized cost of energy of 145 \$/MWh, and a zero carbon dioxide emissions throughout the power generation.

On another hand, the permanence of the supplied electrical power is particularly important in household in village regions and services should be capable of running continuously. Particularly, failure electricity transmission and increased user load demand can lead to a blackout that requires to be prevented. A common idea is to use a backup consisting of a diesel generator and fuel cell. This known solution may be considered as an ideal one. As reported in a multiple studies (Bezmalinovic, Barbir, and Tolj 2013; Braga et al. 2014), hybrid systems containing fuel cell for hydrogen production is becoming financially competitive compared to hybrid renewable systems including only diesel generator as backup systems.

For instance, a study carried out by Aziz et al. (2019) examined the possibility of a combined dispatch control strategy for a hybrid system composed of PV panels with a diesel generator and a battery for backup by combining the cycle charging strategy and the load following strategy using HOMER for simulations and techno-economic analysis. It is revealed that the cycle charging strategy has a net present cost of 110,191 USD and a cost of energy of 0.21 USD/kWh, which are lower of, respectively, 4.8% and 20.6% than values relative to hybrid systems using the combined load following and cycle-charging strategies. Furthermore, the cycle charging strategy provides a best production performance, with less dioxide emissions. In the investigation of Zhang et al. (2018) is proposed an optimal active power control for real-timely by varying thermostatically controlled loads and the functioning points of distributed energy resources and in an islanded microgrid with distributed energy resources consisting of Diesel generator, a microturbine and fuel cell. The authors integrated a large number of thermostatically controlled loads as a load aggregator for participating the secondary control of

load frequency control, which may strengthen the dynamic response performance given their much faster response speeds compared with that of distributed generators. Chapman et al. (2019) reported a review in the aim of assessing the future industrial potential for hydrogen generation by producing top-level system models established on power requirements for enduse services in the United States and United Kingdom with four case studies. For all case models, important potential for hydrogen generation and net energetic benefit are observed. For instance, the authors stated that using hydrogen for on-road transportation in the American state may enable a decline of almost 10% in rejected energy. Besides, for the United Kingdom case, hydrogen could provide the minimum cost alternative for decarbonizing the space heating.

Based on research gap and the literature review discussed above, the main objective of this study is to design and test the performance of a standalone hybrid renewable energy system integrating wind turbine, PV array, fuel cell, and diesel generator system as backup to meet the load demand of a residential community located in Sidi Daoud village in Tunisia. The hybrid system performance is simulated using the TRNSYS software and optimization step with a techno-economic analysis is carried out thereafter while ensuring the supplying of the desirable energy with the best energy performance and lowest investment cost. The proposed HRES system not only must meet the load energy consumption, but also should increase the penetration of renewable energy by presenting a good renewable fraction, a low dumped and unmet power. For the financial and environmental performance point of view, the proposed system should also reach a low-levelized cost of energy and less carbon dioxide emission.

2. Study area and estimated load demand

In the current study, the parameters of the HRES components as the wind turbines, the PV system, the Diesel Engine Generator Set (DEGS), and fuel cell power systems are established on the residential community loads. The considered load consists of an off-grid settlement in a residential community situated in Sidi Daoud village belonging to Nabeul Governorate where a wind farm is established situated in north-eastern of Tunisia. The Sidi Daoud wind farm is a power plant built on a fairly windy coast at the northern tip called "Cap Bon" in Tunisia. It was created and put into service by the Tunisian Electricity and Gas Company (STEG) in 2000 (Bennasr and Verdeil 2012), about 50 km from the capital Tunis. Made up of around 40 wind turbines, and supplying 2.5% of the country's consumption, it was the only one of this type to be in service in 2009 (Ben Amar, Elamouri, and Dhifaoui 2008). The study area is classified as a mild and sunny Mediterranean type climate with the following location parameters: a latitude: $37^{\circ} 1' 44.4''$, a longitude: $10^{\circ} 56' 13.4''$ and a Geodetic system: WGS84. The annual rainfall is around 591 mm, the monthly average temperature varies between 12°C in winter and 27°C in summer. The prevailing winds are from west to west northwest. In addition, the residential community has about 3500 inhabitants occupying 600 houses. The average daily energy consumption per house is 30 kWh/day meaning that the energy demand of the settlement is approximately

18,000 kWh/day. The hourly load throughout the year and the daily profile for a January day as an example are summarized in Figure 1. As observed, a peak demand is obtained in winter months at which further energy consumption are needed due to heating process and less solar radiation availability. In Table 1 are listed the household electrical appliances for a typical house in Sidi Daoud taken into consideration in making the load profile of the settlement.

3. TRNSYS simulation inputs and methodology

The current study used the well-known TRNSYS software for design and simulation of the HRES (Beckman et al. 1994; Buonomano et al. 2018; Chargui, Sammouda, and Farhat 2012; Ulleberg and Mørner 1997). It is a computational tool that is notably used to simulate thermal solar systems with other types of renewable energies like the wind, photovoltaic, and hydrogen systems. Thanks to a dedicated module called Types, TRNSYS also allows energy simulation of buildings.

3.1. Solar potential and PV system modeling

The solar energy amount available on the surface area is considered as the basic-needed information when planning a solar energy transformation system. From this point of view, we take into account the overall amount of solar energy reaching a specific unit surface area (radiation energy and irradiation) in a certain time interval (hourly, daily, monthly, and yearly) rather than computing the instantaneous values. This considered amount of energy is influenced by the season, the orientation, the surface inclination, the meteorological situations, as well as the geographic latitude. As solar radiation depends on multiple effects, estimating the radiation energy by only considering theoretical empirical equations would be insufficient. Therefore, these impacts need to be predicted by examining the measurements achieved from the long-term estimations.

The surface location that is exposed to sunrays remains essential in the solar energy application. A more accurate calculation of the solar energy available in Sidi Daoud region and its possible application are obtained by means of the well-known synthetic weather package one of the most shared synthetic weather data called TMY (Typical Meteorological Years) commonly used sequences used in renewable system simulation. The TMY data gather hourly measurements of weather conditions in assumed meteorological locations to compute all the required data for further analysis, as well as the effect of orientation and inclination change of an assumed surface.

The computation precision extremely depends on the intensity of solar radiation, the air velocity, and the temperature variation of the surrounding air. Consequently, it is compulsory to measure a specific system on an hourly basis for a more comprehensive system calculation. In Figure 2 is illustrated the estimated hourly intensity variation of the total solar radiation for a slope 28 degrees of inclined surface throughout the typical meteorological year TMY for Sidi Daoud village. It is to mention that the TMY also collects certain weather data for a specified geographical site that are selected from hourly

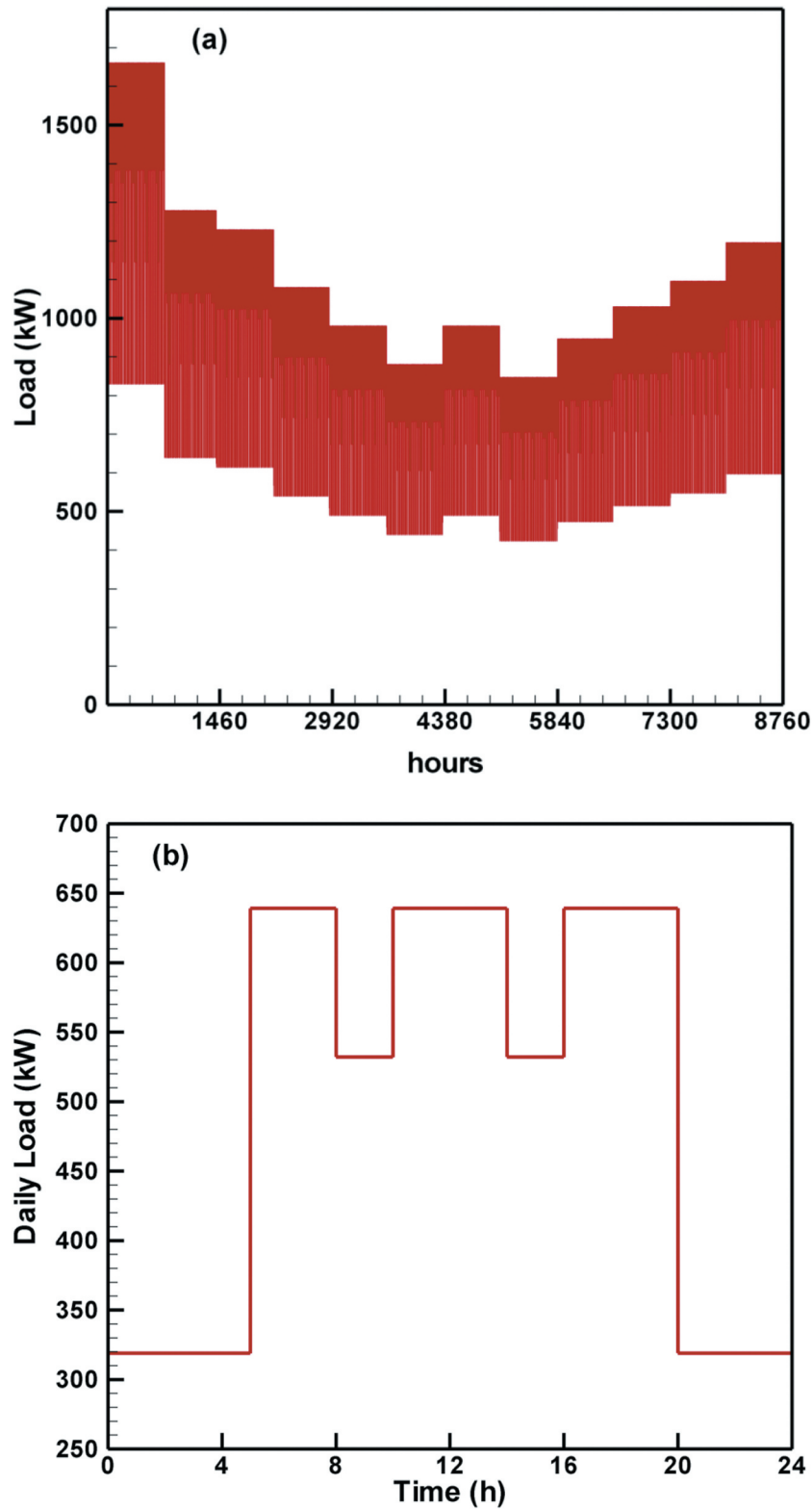


Figure 1. (a) Yearly load for 8760 h of simulations and (b) daily load for the considered residential community in Sidi Daoud region.

data in a longer time period that reaches 10 years or more. It is particularly generated to predict the weather phenomena range of the selected site. Meanwhile, the TMY offers also the annual averages which are consistent with the long-term averages for the same adopted location.

In the current investigation the numerical predictions are carried out using version 16 of TRNSYS software which

involves multiple models to evaluate the solar energy on sloped surfaces.

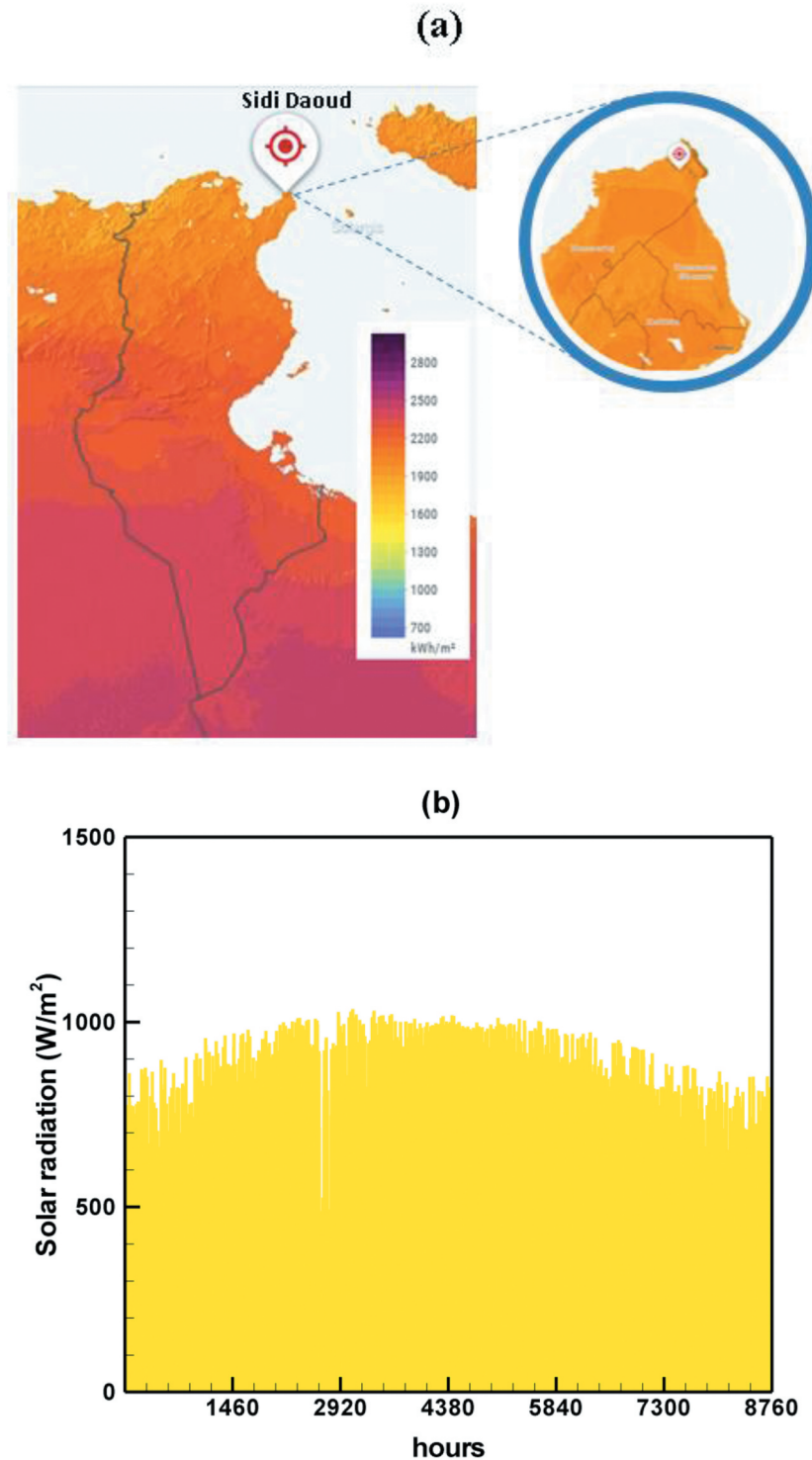
It is worth noting that the optimal tilt angle of the surface during the months of summer (May, June, July) is 28° (fixed based on the study of Jacobson and Jadhav 2018) for which the numerical simulations for PV panels are conducted. The solar energy available on the sloped surface was used for valuing the

Table 1. Electrical appliances relative to the typical house forming the user load demand.

Household Appliances	Day fraction of the appliance used	Appliance power (W)	Average consumed power (W)
Refrigerator	100%	200	200
Air conditioner	25%	2000	500
Desktop computer	50%	160	80
Water heater	60%	400	240
4 Incandescent lamps (4 * 100 W)	50%	400	200
Television	20%	150	30
Average power used in the typical house			1250

electrical energy amount which can be produced by the PV module. In this study, the model SHT72P6 is selected with maximum power 300 W. Its operating temperature is variable between -45 and 85°C . The Short-circuit current of PV-array is 6.88 A while the temperature coefficient is $+0.0555\%/^{\circ}\text{C}$.

In TRNSYS, the PV panel has been simulated using the Type 180 which is a mathematical model for photovoltaic generator. Duffie and Beckman (1991) defined the electrical model while a dynamic thermal model was also described in Ref. (Ulleberg 1998; Ulleberg and Mørner

**Figure 2.** (a) Solar potential and (b) hourly solar radiation in Tunisia and specifically in Sidi Daoud region throughout the year.

1997). The modeling of PV system, also identified as the 5-parameter model, consists of an equivalent circuit of a one-diode model. Besides, despite the model is mainly proposed for PV-arrays containing silicon cells, it can be also used for different materials. In practice, for the configuration of Type 180, the user specifies the number of modules in series in the array, the cells number in series per module, and the number of modules in parallel per array. A Typical plot of current-voltage variation in the PV cell for the adopted photovoltaic model is depicted in Figure 3.

The equation relating the cell voltage U_{cell} and the current I_{cell} of the equivalent circuit is given by:

$$I_{cell} = I_L - I_D - I_{sh} \quad (1)$$

Where I_L is the light current, I_D is the diode current, and I_{sh} denotes the shunt current expressed as follows:

$$I_L = \frac{G_T}{G_{T,ref}} (I_{L,ref} + \mu_{Isc} (T_{cell} - T_{cell,ref})) \quad (2)$$

$$I_D = I_0 \left[\exp \left(\frac{U_{cell} + I_{cell} R_s}{a} \right) - 1 \right] \quad (3)$$

The power for each PV cell is

$$P_{cell} = I_{cell} \cdot U_{cell} \quad (4)$$

Associating these terms to the operation current leads to the array power equation:

$$P_{tot, PV} = N_s \cdot I_{cell} \cdot U_{cell} \quad (5)$$

The measure of PV cell efficiency defined as the ratio of the electric energy produced over the solar radiation flux on a surface which is equal to $G_T = 1000 \text{ W/m}^2$ is finally given by:

$$\eta_{PV, cell} = \frac{P_{cell}}{A_{cell} \times G_T} \quad (6)$$

3.2. Wind energy and turbine model

Tunisia's potential in renewable energy is important and the country has a large wind field with a high wind potential especially in the area of Sidi Daoud belonging to the "Cap Bon" region. Indeed, Tunisia has potential estimated at around 4,000 MW according to the strategic development of renewable energies study carried out by STEG. Besides, the country has opted since 2000 for the development of wind power for production of electricity. Evidenced by this option, STEG is hoping to achieve the 55 MW wind farm production in Sidi Daoud. The wind studies carried out by the company confirms that there exist good wind conditions (with a speed greater than 7 m/s at 60 m high) in this region. Figure 4 depicts the hourly intensity variation of wind velocity in Sidi Daoud for TMY prediction during a year.

On another hand, for TRNSYS simulation, Type 90 is adopted in the current investigation to model the wind turbine by computing the power produced by turbine on the premise of a power versus the characteristic of wind speed in the village of Sidi Daoud. The model type of wind turbine considered in the current study is Made - Endesa AE-52 that is available in the wind farm installed in Sidi Daoud wind farm. The rated capacity of the considered wind turbine is 800 kW, the hub height is 50 m, the rotor diameter 52 m, the cut-in wind speed 3.5 m/s, the cutout wind speed 25.0 m/s, and the rated wind speed is 12 m/s.

The equation describing the calculated power output P_{wind} of the wind turbine by the power coefficient c_p , multiplied by the area of the rotor and power in the wind is given by:

$$P_{wind} = \rho c_p A_r v^3 = \rho 4a(1-a)^2 A_r v^3 \quad (7)$$

Here, ρ denotes the air density (kg/m^3), v is the wind speed (m/s) and A_r is the rotor area (m^2). The coefficient c_p is a function of the axial induction factor a . Its maximum value 59.3% named Betz's limit is the maximum power coefficient

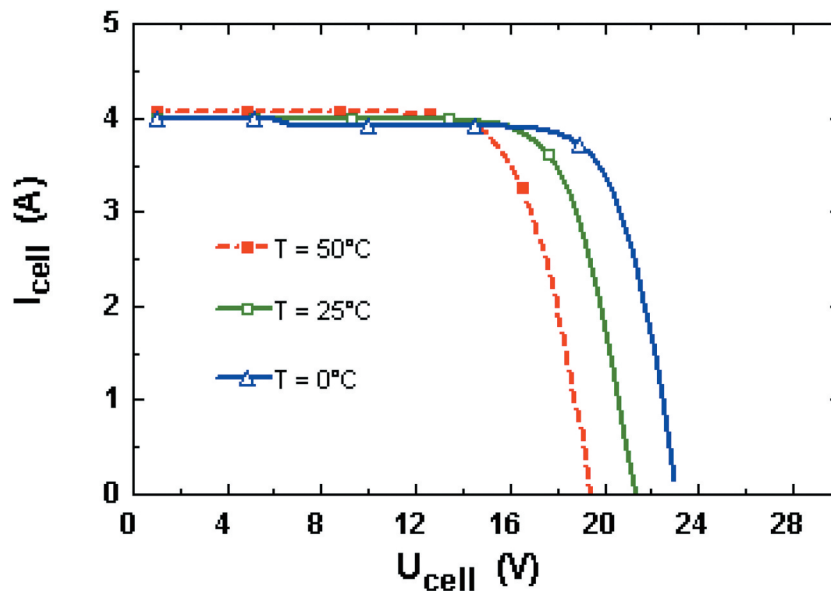
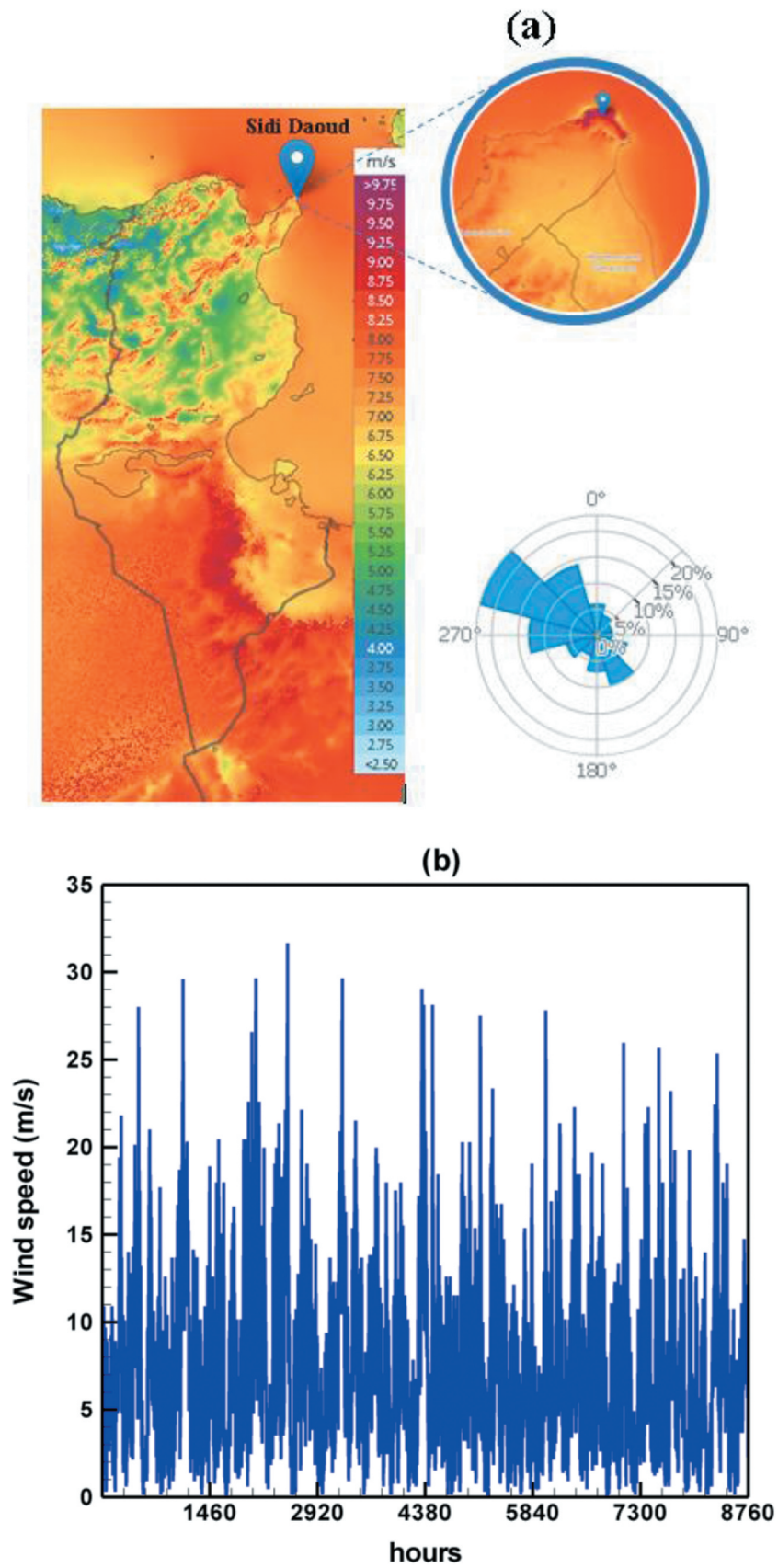


Figure 3. Typical plot of I-U variation for the adopted photovoltaic model.



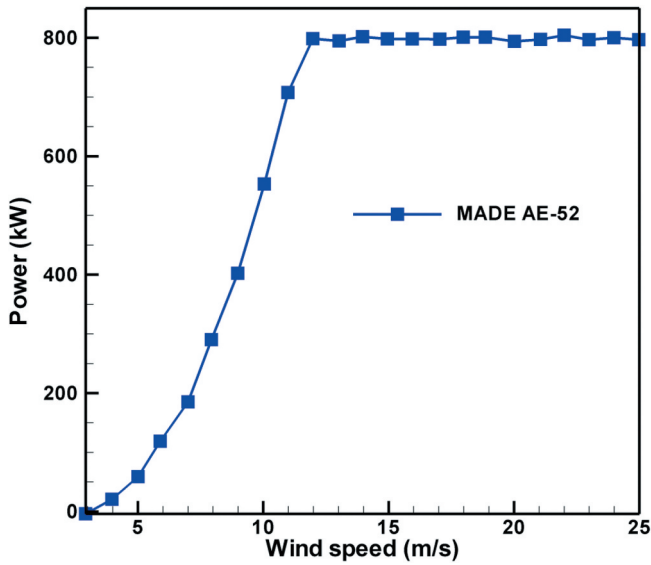


Figure 5. Variation of turbine power with respect to wind speed characteristic for the model AE-52.

that was first derived by Betz in 1919. It is worth noting that the computation of wind turbine power is established on a power variation with respect to wind speed characteristic (Figure 5), in an external file provided based on TMY data. The wind turbine model considers the air density variations and wind speed rises with site elevation. According to the ideal gas law and at a certain height above the ground, the air density varies versus the temperature and pressure as following:

$$\rho_{elev} = \frac{P_{elev}}{RT} \quad (8)$$

According to White (1994), the temperature is reduced linearly with altitude for heights where wind energy applications apply. The temperature “lapse rate” is given by:

$$T(z) \approx T_0 - Bz \quad (9)$$

Here, $B = 6.5$ K/km of altitude and $T_0 = 288$ K.

Based on the theoretical work of Koepl (1982), the equation describing the relation between the height above the ground and the variation of wind speed is:

$$\frac{v_1}{v_2} = \left(\frac{z_1}{z_2} \right)^\alpha \quad (10)$$

The value of the site shear exponent α is almost 0.14 for ideal boundary layer conditions. Nevertheless, under the actual conditions of the region of Sidi Daoud considered the value of α depends on a variety of factors and is significantly higher than 0.14. In this study, the average wind speeds are obtained for $\alpha = 0.16$.

3.3. Diesel engine generator set model

TRNSYS models the diesel engine generator set (DEGS) using the Type 120 which is a mathematical model based on an empirical equation for the fuel consumption given as a function of the normalized electrical power output.

The subroutine of Type 120 allows predicting the required number of operating DEGS units and calculating the performance and the fuel and electrical efficiencies. The default fuel in the current study is natural gas.

The total power output P_{total} is computed by multiplying the electrical power output power from a single engine P_{DEGS} by the number of identical DEGS units as following:

$$P_{total, DEGS} = N_{DEGS} \times P_{DEGS} \quad (11)$$

Another parameter is considered for the DEGS modeling which is the normalized power defined as $X = P_{DEGS} / P_{DEGS, rated}$; where $P_{DEGS, rated}$ is the rated electrical power of engines. This leads to evaluate the fuel consumption given as a linear curve fit as a function of X :

$$\dot{V}_{diesel} = \lambda + \gamma \cdot X \quad (12)$$

Here are λ and γ constants deduced from the linear curve fitting.

Therefore, the electrical efficiency η_{el} is computed as function of the fuel density ρ_{diesel} (kg/m^3), the fuel consumption $\dot{V}_{diesel}$ (m^3/s) and the lower heating value of the fuel LHV_{diesel} (J/kg) as following:

$$\eta_{el} = \frac{P_{DEGS}}{\rho_{diesel} \times \dot{V}_{diesel} \times LHV_{diesel}} \quad (13)$$

The fuel efficiency is hence computed as:

$$\eta_{fuel} = \frac{P_{DEGS}}{\dot{V}_{diesel}} \quad (14)$$

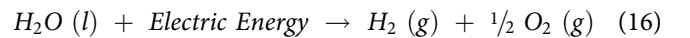
And the total fuel consumption is then calculated as:

$$\dot{V}_{total} = N_{DEGS} \times \dot{V}_{diesel} \quad (15)$$

3.4. Electrolyzer model

In this study, Type 160 is utilized in modeling an advanced alkaline electrolyzer chosen thanks to its reliability, fairly high operational life, and common usage. The water decomposition of H_2 and O_2 can be realized by passing a DC electric current between both electrolyzer electrodes separated by aqueous electrolyte that has good ionic conductivity.

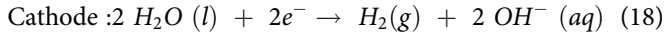
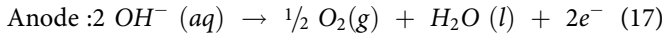
The total reaction describing the splitting water process is:



A minimum electric voltage, or reversible voltage, must be applied to both electrodes for this reaction. This minimum voltage can be determined from Gibbs energy for water decomposition.

In the alkaline electrolyzer considered the electrolyte is generally aqueous potassium hydroxide (KOH), in which the potassium ion (K^+) and hydroxide ion (OH^-) are responsible for the ionic transport. The produced hydrogen is compressed and stored in the hydrogen storage tank to be utilized in the power generation by the fuel cell.

The anodic and cathodic reactions taking place are given by:



The total consumed power P_{ELY} by the alkaline electrolyzer is computed as:

$$P_{\text{ELY}} = \frac{\dot{m}_{\text{H}_2} \text{HHV}_{\text{H}_2}}{\eta_{\text{ELY}}} \quad (19)$$

Here $\dot{m}_{\text{H}_2}$, HHV_{H_2} and η_{ELY} denotes the mass flow rate of the H_2 produced (kg/s), the heating value of the hydrogen fuel (MJ/kg), and the electrolyzer efficiency, respectively.

The electrode kinetics of the electrolyzer cell is modeled by means of the empirical (I-U) current-voltage relationships. The ohmic resistance, over voltages, and temperature must be considered and related using the following equation:

$$U_{\text{cell,ely}} = U_{\text{rev}} + \frac{r_1 + r_2 T_{\text{ely}}}{A_{\text{electrolyte}}} I_{\text{ely}} + (s_1 + s_2 T_{\text{ely}} + s_3) T_{\text{ely}}^2 \log \left(\frac{t_1 + t_2 T_{\text{ely}} + t_3 T_{\text{ely}}^2}{A_{\text{electrolyte}}} I_{\text{ely}} + 1 \right) \quad (20)$$

Here, $U_{\text{cell,ely}}$ is the operation cell voltage (V), U_{rev} is the reversible cell voltage (V), I_{ely} is the current through cell (A), r_i ($i = 1, \dots, 2$) parameters for Ohmic resistance of electrolyte, s_i , t_i ($i = 1, \dots, 3$) are parameters for overvoltage on electrodes and $A_{\text{electrolyte}}$ (m^2) and T_{ely} ($^{\circ}\text{C}$) are the area and temperature of the electrolyte, respectively. Note that, the empirical parameters (r_i , s_i and t_i) can be numerically computed using non-linear regression techniques.

In addition, according to Faraday's law, the production rate of hydrogen ($\dot{n}_{\text{H}_2}$) in the electrolyzer cell that is directly proportional to the transfer rate of electrons at the electrodes, which in turn is equivalent to the electrical current in the external circuit. Therefore, we may express the total hydrogen production rate ($\dot{n}_{\text{H}_2}$) in an electrolyzer consisting of several cells connected in series as follows:

$$\dot{n}_{\text{H}_2} = \eta_F \frac{n_c I_{\text{ely}}}{nF} \quad (21)$$

Where, η_F is the Faraday efficiency, n_c the number of cells in series, n is the number of moles of electrons per moles of water ($n = 2$), and F denotes the Faraday constant ($F = 96,485 \text{ C/mol}$). The parameter η_F is expressed as the ratio between the actual and theoretical maximum quantity of hydrogen produced in the electrolyzer.

Consequently, the produced hydrogen by the electrolyzer is compressed and stored in the hydrogen tank simulated via the Type 164 in TRNSYS. This type performs simply moles or mass balance of the hydrogen gas that enters and leaves the storage and computes the pressure resultant from mass of Hydrogen in the tank. In case the pressure augments beyond a fixed level, the H_2 excess is dumped.

3.5. Description of fuel cell model

The fuel cell is modeled using Type 173 in TRNSYS relative to an alkaline fuel cell (AFC) which is a device allowing to convert a chemical energy of Hydrogen as a fuel cell and the air as an oxidant to DC electrical current. The electrochemical model is established on an empirical relationship relative to the characteristic of current-voltage at the normal operating temperature (Figure 6).

The fuel cell voltage U_{cell} expressed as $U_{\text{cell}} = U_{\text{mod}}/n_{c,ser}$ is the module voltage U_{mod} divided by the number of cells in series $n_{c,ser}$. The voltage across each AFC stack U_{stack} can be deduced by multiplying the voltage module U_{mod} by the number of AFC modules in series per stack $n_{m,ser}$ as follows:

$$U_{\text{stack}} = n_{m,ser} \cdot U_{\text{mod}} \quad (22)$$

Where

$$U_{\text{mod}} = U_0 - b_{\text{Tafel}} \cdot \log(I_{\text{stack}}) - R_{\text{ohm}} \cdot I_{\text{stack}} \quad (23)$$

The parameters U_0 is the open circuit voltage, R_{ohm} denotes the Ohmic resistance per AFC module basis and b_{Tafel} is the Tafel slope, respectively.

Likewise, the stack current given by $I_{\text{stack}} = I_{\text{FC}}/n_{s,par}$ can be obtained from the cell current expressed as $I_{\text{cell}} = (I_{\text{FC}}/(n_{c,par} \cdot n_{s,par}))$. Here, I_{FC} is the total electrical current provided by the fuel cell unit, and $n_{c,par}$ and $n_{s,par}$ refer to

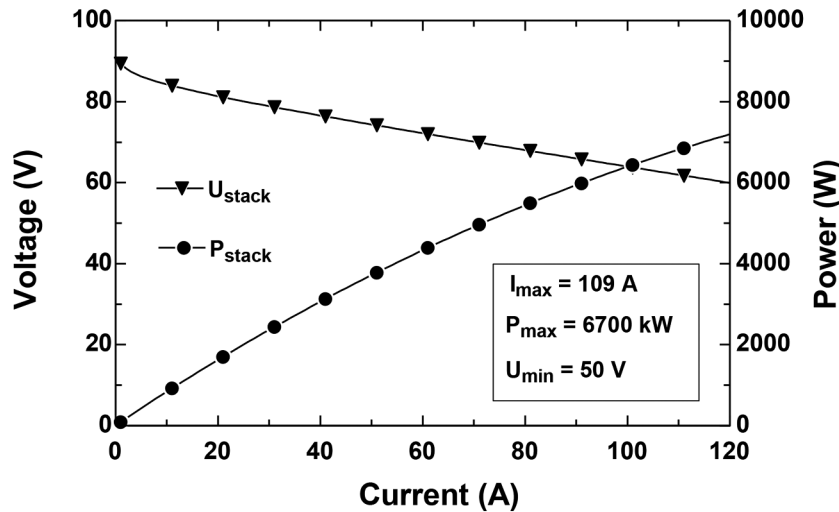


Figure 6. Current-Voltage curve relative to the AFC stack model.

the number of cells and stacks in parallel per AFC unit, respectively.

The stack power output can be thus found as:

$$P_{stack} = U_{stack} \cdot I_{stack} \quad (24)$$

The energy efficiency η_E is obtained in terms of the cell voltage U_{cell} , electrons number n_e , the Faraday constant F and the enthalpy (ΔH_2) for hydrogen at standard conditions:

$$\eta_E = \frac{U_{cell} \cdot n_e \cdot F}{\Delta H_2} \quad (25)$$

3.6. Power conditioning unit model

In TRNSYS, the Type 175 is used to mathematically simulate the power conditioning unit based on empirical efficiency curves for inverters (DC/AC or AC/DC) or electrical converters (DC/DC). In our stand alone power system involving wind turbine and PV array, the system input power is continuously variable with time. The output characteristics of a wind turbine and PV array have peak power points depending on wind speeds, cell temperature, and solar insolation, respectively. Hence, it may be helpful to use a maximum power point tracker (MPPT) to use the input power source to the fullest capability. The power loss (P_{loss}) for the converter depends on the electrical current running through it and is described by the three-parameter expression given as:

$$\begin{aligned} P_{loss} &= P_{in} - P_{out} \\ &= P_0 + (U_s/U_{out}) P_{out} + (R_{ipn} / U_{out}^2) P_{out}^2 \end{aligned} \quad (26)$$

Where P_{in} , P_{out} , P_0 , U_s , U_{out} , R_{ipn} are the inlet and output power of the conditioner, the output and set-point voltage and the internal resistance constant, respectively.

The electric efficiency for the converters is:

$$\eta_{conv} = \frac{P_{out}}{P_{in}} \quad (27)$$

4. Optimization and control strategies

4.1. System design and economic optimization process

The design of our HRES system in Sidi Daoud village location with its known insolation and wind power is dependent tremendously on the load demand of the houses. Some of the most viable alternative designs are the direct coupling of the wind turbine, PV-arrays, DEGS, the electrolyzer, and the fuel cell to the DC-busbar. Besides, the optimization process requires replacing the low-pressure electrolyzer with a high-pressure electrolyzer. Furthermore, for our system involving the wind turbines and load, a wind energy conversion system will influence the hybrid system design in two major techniques; first, by diminishing the required PV array area, and second by reducing the required Hydrogen storage volume to get a lower cost of the total system.

Considering the methodology of designing the most power-efficient and practical PV-wind-hydrogen-DEGS system the following steps are adopted:

1. The expected wind energy, solar radiation, and load demand for the residential site in question must be determined from annual term data in TMY file.
2. A system configuration is firstly selected.
3. Once a feasible system configuration has been established, a detailed analysis of the system operation must be performed.
4. A systematic methodology on modeling, testing and verification of the proposed model, and TRNSYS simulation are carried out.
5. If step 4 shows that it is possible to reduce the sizes of the initially selected components with the same or better energy power, a redesign of the HRES configuration must be executed.
6. A combined optimization of the design and operation (steps 3–5) is then imposed.

During these 6 power and designing steps, we carried out a space optimization investigation relatively to each component (wind turbine, PV panel, the fuel cell, the DEGS, electrolyzer, the hydrogen tank, and the converters) based on the size of every element and a power dispatch control mode realized by a controller. The optimization investigation of the HRES design has been performed by varying the output power capacity of the wind turbine from 0 to 1600 W, the PV panel from 0 to 2000 kW, the fuel cell up to 600 kW of output. The inlet power capacity of the electrolyzer has been also tested until 600 W. In addition, the capacity of the Hydrogen tank is varied up to 900 kg. The converters' upper limit power is varied from 0 to 800 W.

In addition, in order to investigate the best solution satisfying the wanted constraints, multiple hourly computations of the considered system configurations have been performed during the year with a total simulation of 8640 hours.

On another hand, in parallel to energy and sizing methodology, an economic optimization examination is carried out based on the preferred constraints at the lowest NPV of total cost to predict the optimum hybrid power system configuration in terms of financing cost. The index of performance is chosen to be the levelized cost of energy (LCOE) calculated in (\$/kWh) and the life-cycle cost. The cost of energy analysis is based on different parameters, namely; the capital cost, replacement cost, operation and maintenance cost, the life cycle, fuel cost, the capacity and the financial revenue gained from electricity generated (Braga et al. 2014). Therefore, in the aim of designing the most economical HRES the following step are required:

1. Go through the aforementioned designing steps 1–5 described above.
2. Define the cost function for the different system components and for several alternatives.
3. Make a combined optimization of the HRES design and operation for each alternative (step 6).
4. Compare and evaluate each (optimized) alternative and meanwhile calculating the LCOE each time.

Finally, the designing and economic optimization examinations lead to the best configuration obtained for a combination of 793 kW wind turbine, 1167 kW PV array, 1500 kW for the 5 diesel engines (300 kW for one single DEGS), a 295 kW fuel cell, a 500 kW electrolyzer, 916 kg hydrogen tank and 296 kW for the converters.

4.2. Convergence solution and stability checking

For the convergence criterion, a function called “*getConvergenceTolerance*” is implemented in TRNSYS to guarantee the solution convergence. It is a double precision function that returns the user specified value of a simulation’s convergence tolerance (the second argument in the tolerances keyword statement).

Moreover, the TRNSYS solvers use a check stability function called “*CheckStability*”. It is an integer function that returns a value 1 if the last time step converged and a 0 if the last time step did not converge. Using the Powell’s method, the *CheckStability* function is used in the solver, allowing more than one control strategy and then backs up a time step to try something else if it did not find a stable solution. TRNSYS data reading components need to know that they should not continue reading the data file but should back up as well. Likewise, TRNSYS uses an *INFO* array that is an integer array of 15 values which conveys to the TRNSYS Types and to the kernel information about the current status of the *UNIT*. Among them is the one called *INFO(13)* which indicates that all the Units have converged during the current time step. It is 0 during iterative calls and 1 during the post-convergence call. On another hand, the relative accuracy of the HRES simulation can be predicted by comparing the energy flows across the system or its components with the variation in HRES internal energy. If the energy flows between several components balance, then the time-step and convergence tolerances are satisfactory. In the opposite case, where an energy balance is not satisfied, then the HRES model or computation parameters have not been specified properly. Hence, to overcome this issue, an energy balance checking routine is included in TRNSYS to simplify the checking procedure. A *CHECK* statement that is a control statement is utilized to identify an energy balance between several quantities printed by the simulation summary component. If the energy balance does not close within the tolerance given on the *CHECK* statement, the simulation is terminated.

4.3. Dispatch control strategies

In the current TRNSYS modeling, the Power Dispatch Control Mode (PDCM) is adopted in the simulation methodology which is composed of two modes of the renewable energy system described in Figure 7(b). The first strategy is the Load Following Mode (LFM) and the second one is the Cycle Charging Mode (CCM). In the LFM, the wind turbine and PV panel are considered the primary load to meet the demand. In this case, the generator stays off and no excess power exists. In case the wind turbine and the PV power are lower than the load power and do not meet the energy demand, the fuel cell will run at a rate providing enough power to meet the load demand, while the generator operates and produce enough power to meet the load. Meanwhile, the diesel engine generator system (DEGS) manifested by Type 105 determines the total number of DEGS and the power per DEGS required, the fuel cell power, and electrolyzer power to meet the given load. Besides, the maximum (rated) and minimum (idling) powers for the fuel cell and electrolyzer are taken into consideration.

On another hand, the Cycle Charging Mode (CCM) is nearly identical to LFM. Nevertheless, the mode differs from the LFM in that when the generator is switched on, it runs at its maximum-rated capacity and the fuel cell will operate at full power rate as well to supply the net load with excess energy. It is worth noting that, the operation is executed by the controller exhibited by Type 105 which contains the control functions for the fuel cell (FC) and the electrolyzer operating with the diesel engine generator system (DEGS). Hysteresis for the fuel cell and electrolyzer on/off-switching is also introduced. Note that ‘off’ in this case actually means that the unit is idling until the hydrogen storage pressure level, or state of charge (SOC), has reached its predefined set point. Based on the above dispatch control modes, the controller provides the current references to discharge the fuel cell and stock the hydrogen through electrolyzer.

5. Modeling results and discussions

5.1. TRNSYS simulations and HRES performance

Results of the TRNSYS simulations and performance of the standalone HRES for the residential community in Sidi Daoud village are analyzed in this section. In Figure 7 is sketched the system schematic of the hybrid system and the diagram illustration used in the TRNSYS software showing the configurations of the renewable hybrid power system proposed herein.

The software is run at 60-min time-steps throughout a full year. In addition, consistency is maintained between the Typical Meteorological Year (TMY) files used for load demand modeling and TMY wind and solar resource data used for modeling.

The considered configuration of the PV/wind/DEGS/AFC system using TRNSYS software are based on the user loads, components technical and cost details, the wind and solar and resources availability. TRNSYS simulates the operation of the HRES by making energy balance computations in each time step (1 h) of the year. The computation results consist of the electricity production from the wind turbine, the PV system diesel engines, and fuel cell. Meanwhile, TRNSYS takes into account the energy consumption by the load demand, the input DC power of the electrolyzer, the power losses of both AC/DC and DC/AC converters as well as the dumped and unmet power. Additionally, numerical predictions allows us to carry out an economical investigation in terms of the cost summary of system components, the operation and maintenance (O&M), the replacement cost, and dioxide carbon emission CO₂ emitted by the HRES. Hourly, daily and monthly average energy production, as well as the cash flow over the 20 years life time and salvage of the system are estimated.

As aforementioned, an optimization analysis based on thousands of test simulations has been firstly performed to predict the best configuration of the HRES taking into account the optimum compromise for lowest present cost and power outputs reliability. It is also worth mentioning that the optimized HRES is also obtained for a cycle charging control strategy.

Considering the multiple simulations and tests performed, we present in details the results relatively to the best configuration obtained in the current section.

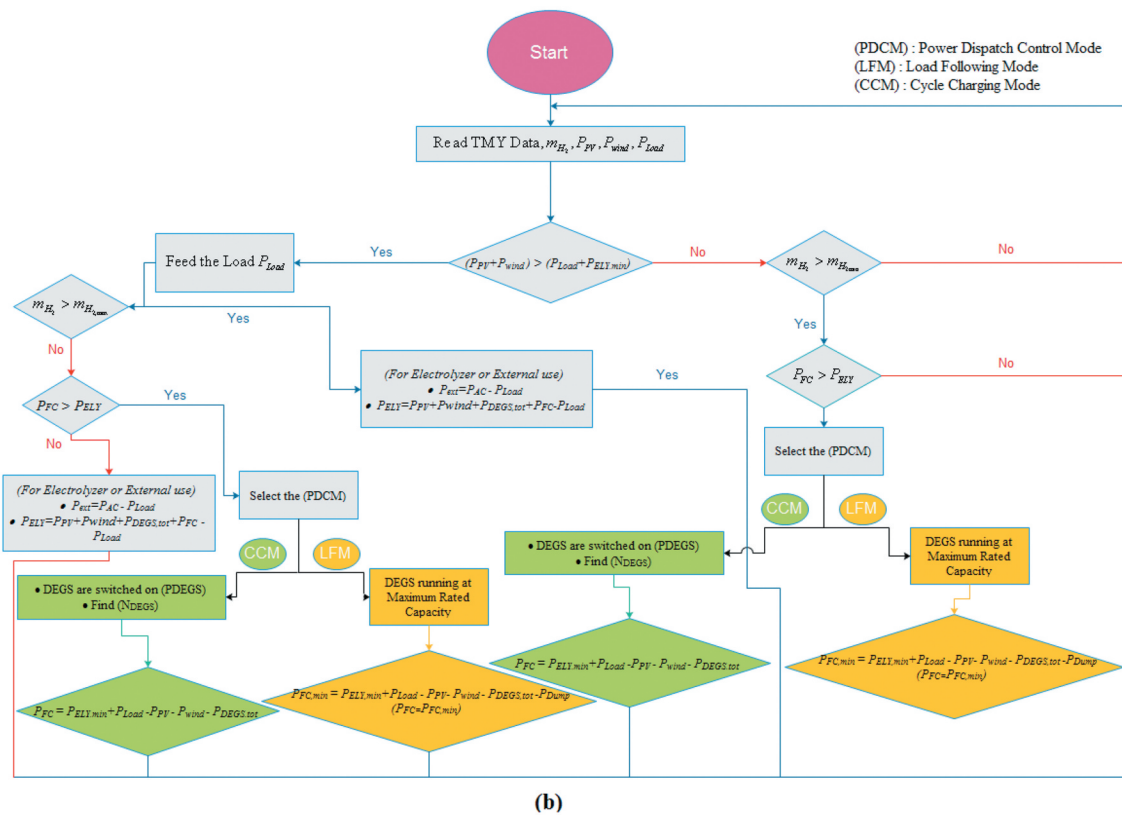
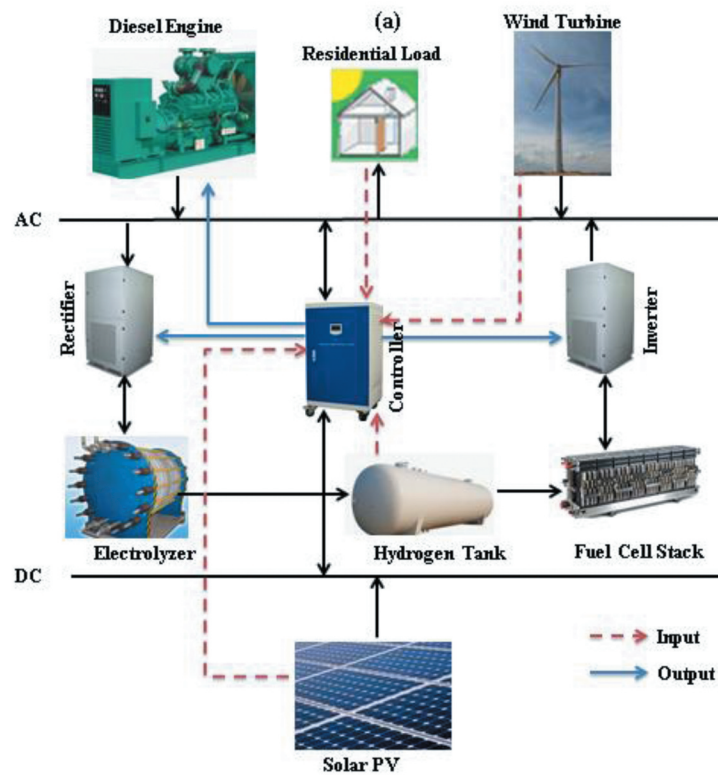


Figure 7. (a) System schematic and (b) proposed power management algorithm of the TRNSYS simulation with LFM and CCM strategies.

The obtained rated power of the wind turbine is 793 kW while an annual mean power output is found to be 263.092 kW. The yearly energy output is 2,304.950 MWh. The capacity factor of the considered wind turbine Made A-52 is estimated

to be 33.187% presenting the ratio of average-delivered power to theoretical maximum power. Furthermore, the maximum average power coefficient (C_p) of the wind turbine system defined as the ratio of the total electrical power produced

divided by the wind power into the turbine is found to be $C_p = 44\%$.

For the PV panel, the annual mean power output is 244.560 kW and an annual energy output of 2142.59 MWh/year is estimated. This presents a mean output of 5869.44 kWh/day. This leads to a capacity factor of the PV system of 20.97% representing the actual annual energy output over time divided by the maximum possible output, while the efficiency of the PV is 13%. For the fuel cell the maximum electrical output is 294.938 kW but a mean electrical power output of 97.331 kW is generated from the alkaline fuel cell. The yearly energy contribution of the fuel cell output is estimated to be 852.722 MWh/year meaning a daily production of 2,236.093 kWh/day with an electrical efficiency of 56.22% for the fuel cell.

Considering the diesel engines system, the rated power is found to be 1.5 MW with an average power output 556.505 W meaning an annual energy production of 4,875.54 MWh/year. The efficiency of the diesel engine system is found to be 40.747%.

Furthermore, the mean power output of the AC/DC converter connected to the electrolyzer is 239.473 kW and a maximum out power obtained 485.003 kW for 8760 h per year of operation. The AC/DC converter inlet energy is about 2,179.18 MWh/year while an out energy is predicted to be 2,097.77 MWh/year. The energy losses are 81,879 kWh/year (0.24% of the total system hybrid production) with an average annual efficiency of 95.074% for AC/DC converter. Considering the DC/AC converter related to the fuel cell we obtained an entering yearly energy in of 876.931 MWh/year and a simulated leaving energy out 861.52 MWh/year with a maximum power out of 98.336 kW. The yearly energy losses from the DC/AC convert are found to be 24,258.9 kWh per year which only

signifies 0.8% of the total production of the hybrid system. The mean efficiency of the DC/AC converter is calculated to be 96.6%.

If we consider the proposed hybrid system, the total electrical energy produced is 10,200 MWh/year with 2,304.95 MWh/year contributed from the wind turbine (22.59%), 2,142.586221 MWh/year from the PV system (21.01%) and 853 MWh/year produced from the Fuel Cell (8.36%) as illustrated in Figure 8. On the other hand, the annual backup energy required and covered by the DEGS is predicted to be 4,875.54 MWh/year which means a percent of 47.79% from total produced energy by the hybrid system.

In addition, the HRES feeds the total users load with energy of 7,560 MWh/year and only less than 0.05% of unmet load is predicted (5.23 MWh/year which is less than 0.015 MWh/day) as shown in the Figure 8. Moreover, the electrolyzer consumes 2,174.9 MWh/year presenting a percentage of 21.32% of the total system hybrid production with a dumped energy of 465.7634 MWh which presents only 4.7% of the total system production. It is also interesting to mention that the total system power losses from converting procedure are only 106.138 MWh/year (1.02% of the total production) which characterizes the energy losses in both the fuel cell and electrolyzer converters during the conversion process.

An interesting parameter depicted in Figure 8 reflecting the capability of the proposed HRES model is the renewable fraction abbreviated as (RF) which defines the energy provided to the load demand that is originated from the renewable power which are the wind, PV and fuel cell sources and calculated as following:

$$RF = (1 - (E_{Aux}/E_{Load})) \quad (28)$$

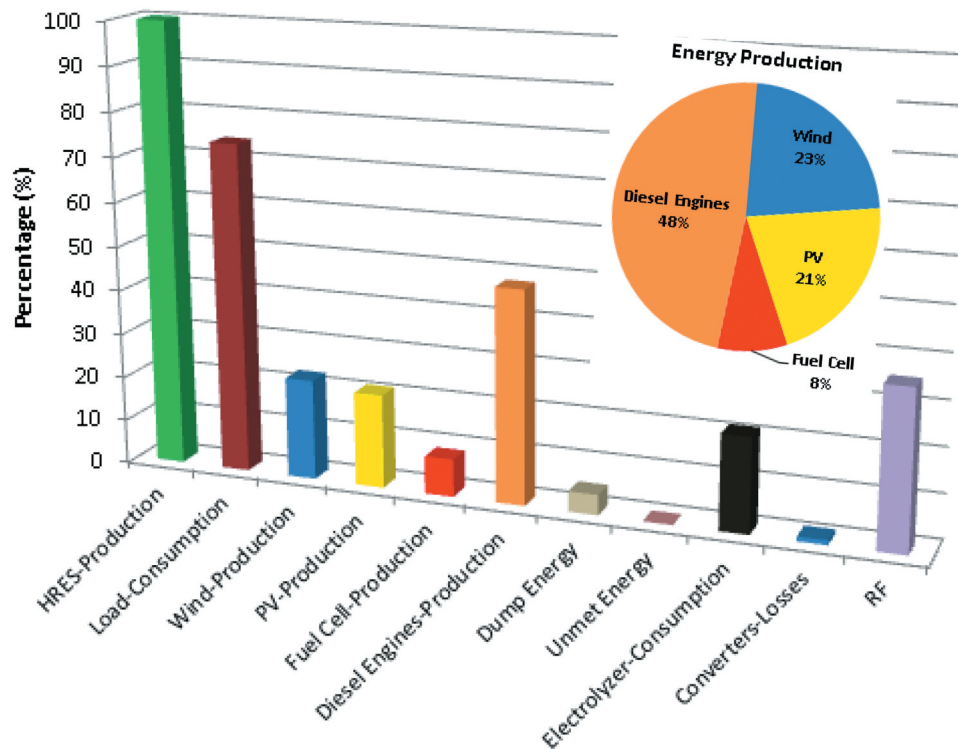


Figure 8. Annual produced, consumed, lost and dumped energies with the renewable fraction of the optimal HRES.

Where E_{Aux} (kWh/yr) denotes the auxiliary nonrenewable electrical source which is the diesel engines and E_{Load} (kWh/yr) is the total electrical load served. The obtained renewable fraction of the proposed hybrid system is $RF = 35.52\%$. Such obtained percentage of Renewable Fraction penetration in this investigation after optimization procedure seems to be acceptable. In fact, the performance of our proposed HRES is examined with multiple evaluation criteria, which include power generation performance, excess and dumped energy, NPV, LCOE, RF, and environmental impact. Therefore, in the optimization sizing and economic process, all these parameters are taken into account involving the Renewable Fraction. Indeed, in order to get the best optimized system the excess and dumped energy must not be elevated and in turns getting an augmented RF with less percentage of DEGS power contribution. On another hand, for comparison with RF present in literature, especially those involving backup components, one can cite the investigation of Halabi and Mekhilef (2018). This investigation dealing with the optimization of multiple hybrid systems for both off-grid and on-grid connections involving photovoltaic modules and two diesel generators, a converter and a battery banks, demonstrated that the different renewable energy fraction (RF) values are obtained in the range

($23\% \leq RF \leq 55.43\%$). Furthermore, Ghenai, Salameh, and Merabet (2020) who studied the solar PV/Fuel Cell off-grid power system integrated with solar-based electrolyzer stated that they obtained a penetration of renewable resources of $RF = 40.2\%$. Moreover, Awan et al. (2019) reported that for the PV-diesel-battery system that is economically the top-ranked system among the other combinations of hybrid systems, this system configuration has a RF of 37.1%. In the study of Aziz et al. (2019) for a photovoltaic (PV)/diesel/battery hybrid system, the renewable fractions for multiple dispatch strategies, namely Load Following, Cycle Charging, and Combined Dispatch strategies were 44.7%, 18.4%, and 35.6%, respectively. In addition, in the Load Following dispatch strategy; Aziz et al. (2019) stated that the diesel generator contribution was approximately 40.6% while a 73.8% share of the total energy production is observed in the Charging Cycle mode.

Figure 9 illustrates the monthly electrical energies and corresponding hourly power produced by the different components of the HRES consisting of the PV system, wind turbine, diesel engines, and fuel cell. It can be seen that the diesel generators are mostly activated in the winter especially in January and December when less sun is available in Tunisia and particularly in Sidi Daoud village. As expected, a more

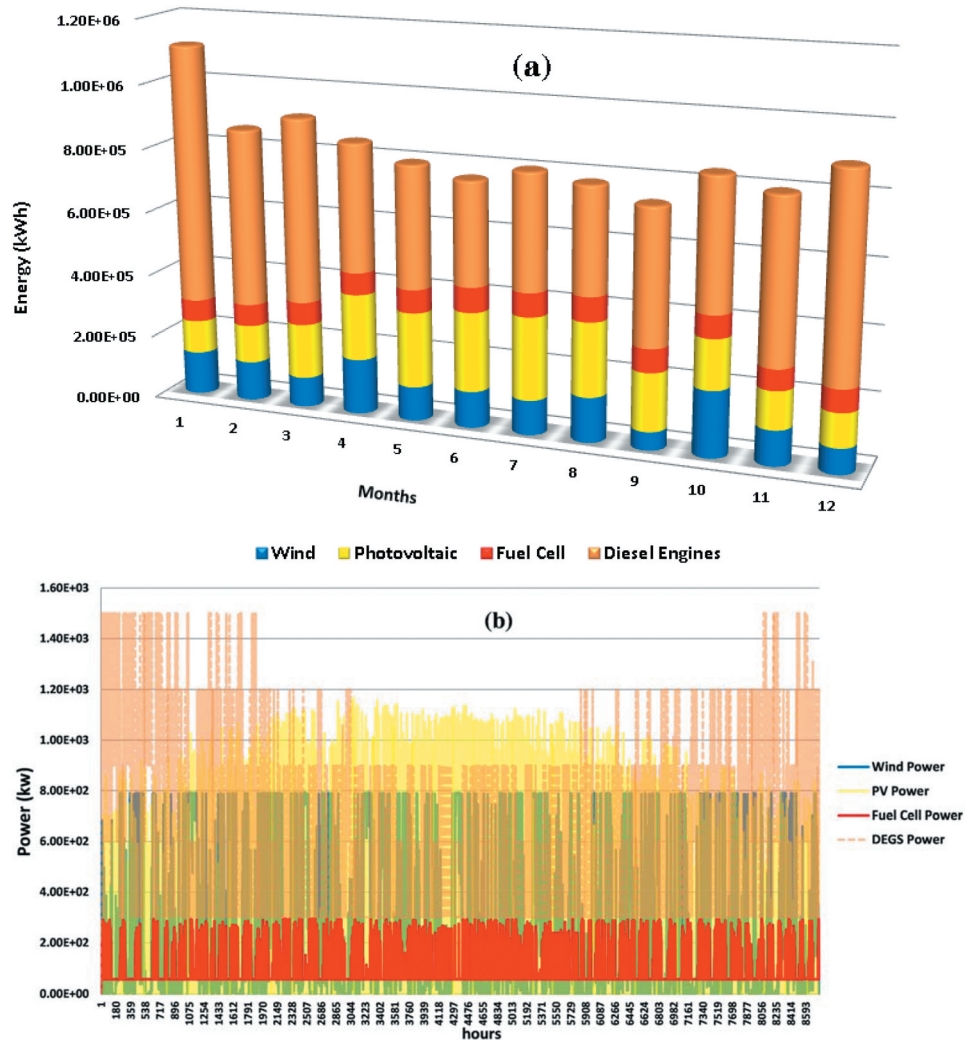


Figure 9. (a) Monthly energies and (b) hourly power produced by each hybrid system components with respect to the optimum HRES.

energy is produced by the PV system in the summer months while the contribution of the wind power is less pronounced in this period of year. Besides, regardless of the load status, the fuel cell is only operating when needed, and hence does not produce excess power unlike the PV system and wind turbine which operate throughout the year.

On the other hand, the monthly and hourly energy consumption, losses, dumped electricity and unmet load of the optimum HRES visualized in Figure 10 clearly demonstrate that the maximum monthly electricity demand reached is in winter periods while the lowest value was during June month in the summer period where the sun availability is at its peak.

In order to get a better understanding of the HRES operation, details of the hybrid system's power load, the unmet load, the dumped electricity, and the amount of generated power by each component during two chosen weeks in winter (15th to 22nd January) and summer (15th to 22nd July) seasons are depicted in Figure 11. The results show that most of the power generation during the day hours is gained from the solar PV, and the DEGS seems to cover the power demand

during night with the fuel cell operating when only required. Hence, peaks of the total HRES production are obtained at noon hours for both winter and summer weeks. Furthermore, most of the primary load demand for the residential community is met by the HRES with a low-dumped power from the HRES that is used for running the electrolyzer. Thanks to the existence of an engine backup in the total system, the unmet energy is nearly zero demonstrating the capability of the renewable energy system to meet the growth of users demand. As noted earlier, a common solution to generate electricity for standalone systems is using a diesel engine to avoid unmet loads. The wind energy at the city of Sidi Daoud is higher during the winter months but does not seem to be dependent on the day hour whether it is during night or day as observed in the figure.

Figure 12 visualizes the hourly volume and pressure of hydrogen in the storage tank throughout the year. It is seen that the storage of H₂ reaches its higher levels in the summer period where the energy consumption is at its peak, which will compensate the needed supply energy storage for the

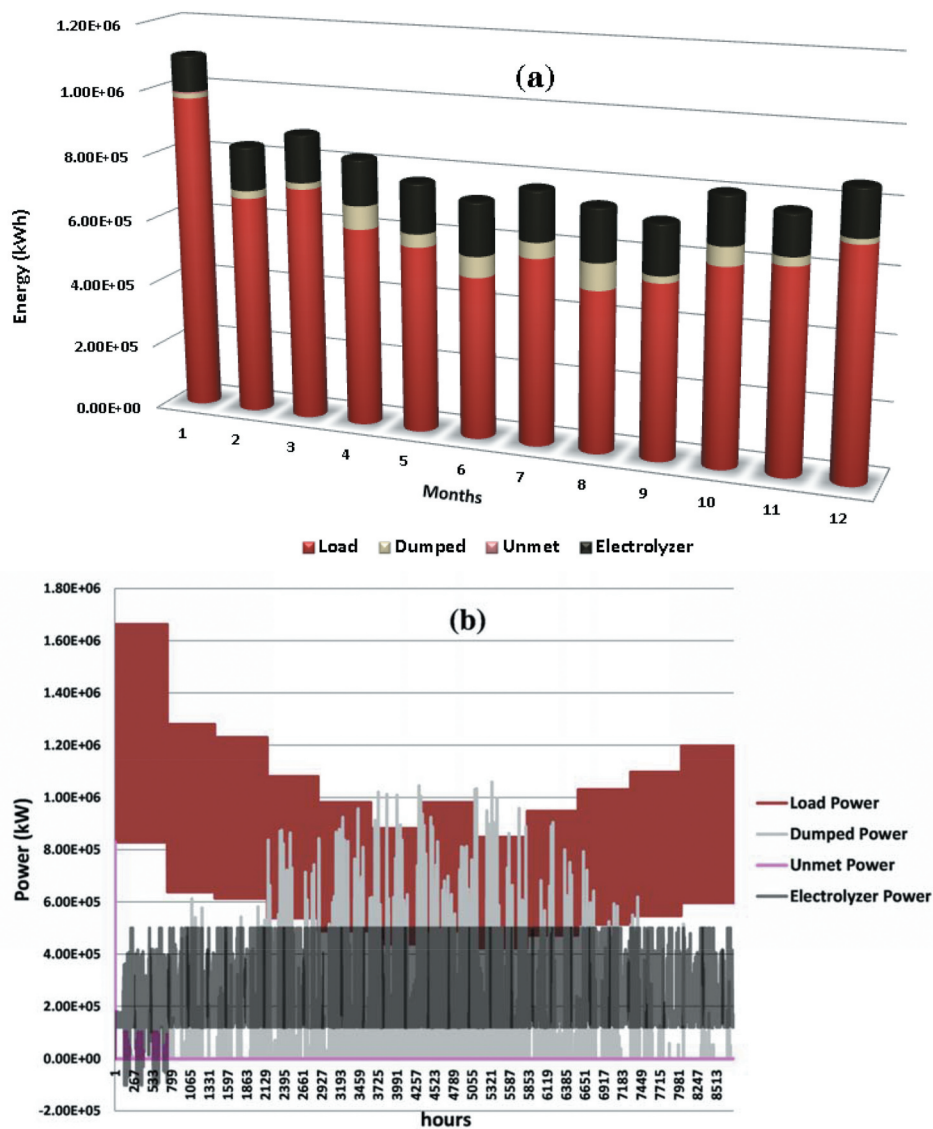


Figure 10. (a) Monthly energy and (b) hourly power consumption, losses, dumped electricity and unmet load of the optimum HRES.

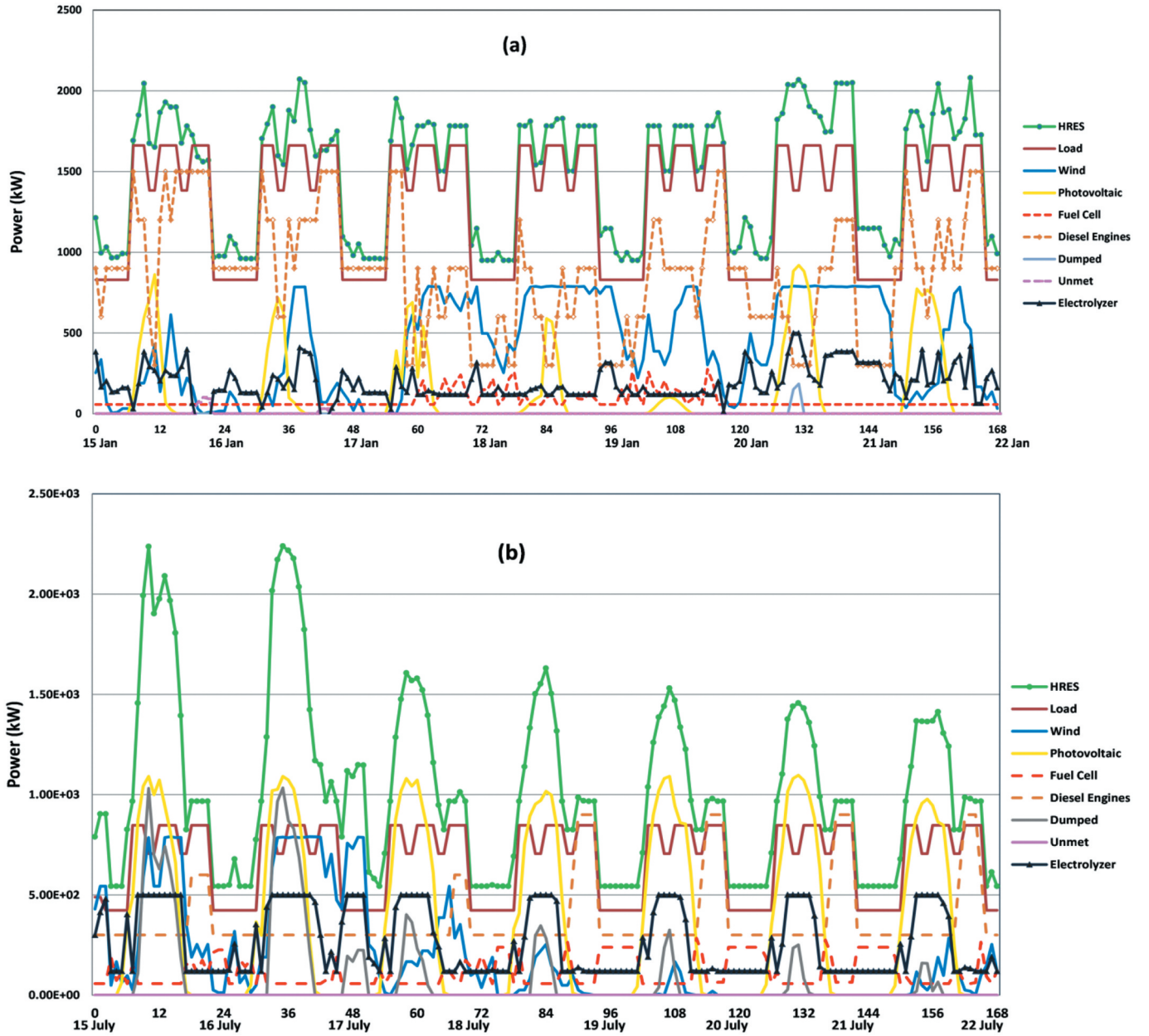


Figure 11. Typical weekly performance of the HRES power during the (a) winter (15th to 22nd January) and (b) summer (15th to 22nd July) seasons.

electrolyzer system in the winter months at which the pressure of hydrogen in the tank diminish drastically.

On another hand, the hourly evolutions of the power output of the Alkaline Fuel cell (P_{FC}) and negative input power of the Electrolyzer ($-P_{ELY}$) relative to the optimal renewable hybrid system throughout the year are also given in Figure 13. In accordance with the previous Figure 12, the closer look at the data in Figure 13 shows that the electrolyzer frequently needed to operate in winter period on grid-electricity to generate extra hydrogen and level out the pressure of hydrogen storage, which otherwise declines rapidly. In fact, the system controller is automatically programmed to switch on the electrolyzer during only time of excess power (PV-Wind-Load). Once the excess energy on the microgrid goes below a critical low point, the electrolyzer is turned off while the diesel engine is switched on to cover the demand.

In order to reveal the stability of the hybrid system and to ensure that the hydrogen storage tank meets the refueling requirements, we use the key indicator of State of Charge (SOC) defined as the density of hydrogen during refueling to the H_2 full-filling density (Figure 14). It is worth noting that the SOC of initial SOC value of the hybrid system is set as 50%. As can be seen from the figure, SOC of the hybrid system in August (summer month) is slightly higher than that in December (winter month) which means that the proposed system can improve his stability especially in hotter periods. In fact, the monthly SOC evolution trend follows the monthly availability pattern of renewable energy sources that are further augmented in summer with further strengthening of photovoltaic power.

An illustration of the variation trend of the dump power (P_{Dump}) and the required number of diesel generation (N_{DEGS}) throughout the year is presented in Figure 15. It

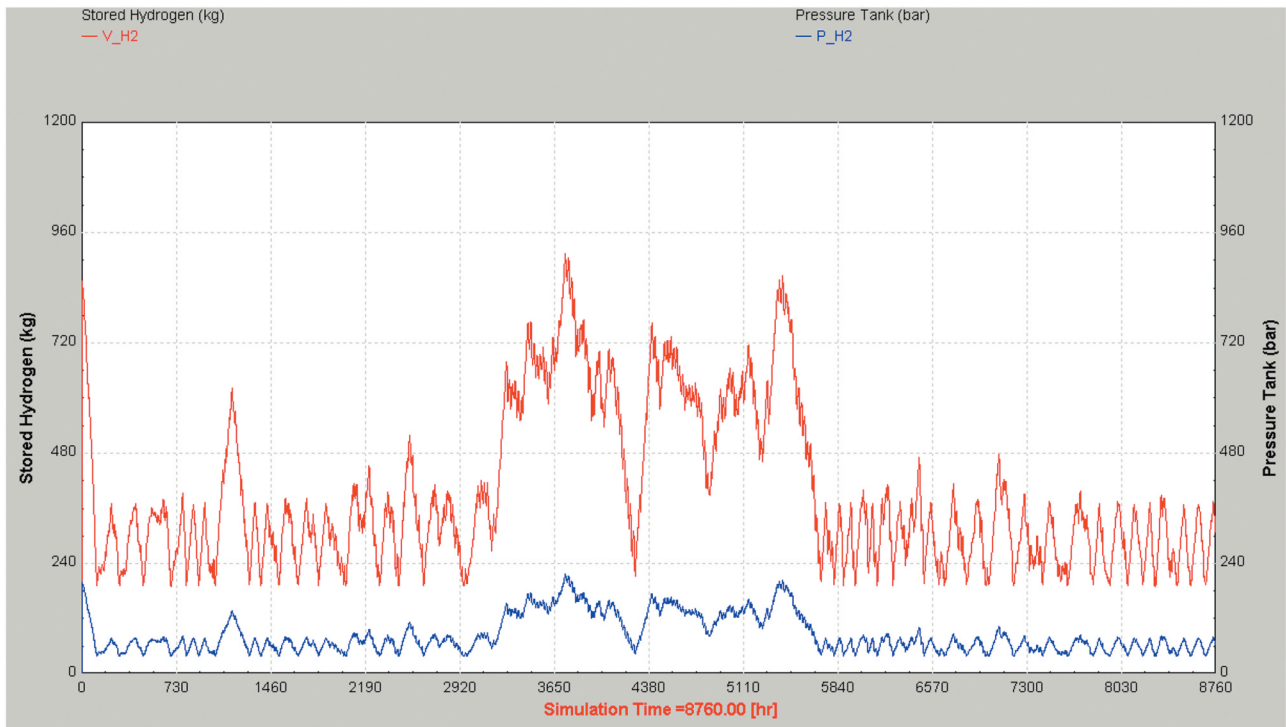


Figure 12. Hourly volume and pressure of Hydrogen in the tank of the optimal HRES during the year.

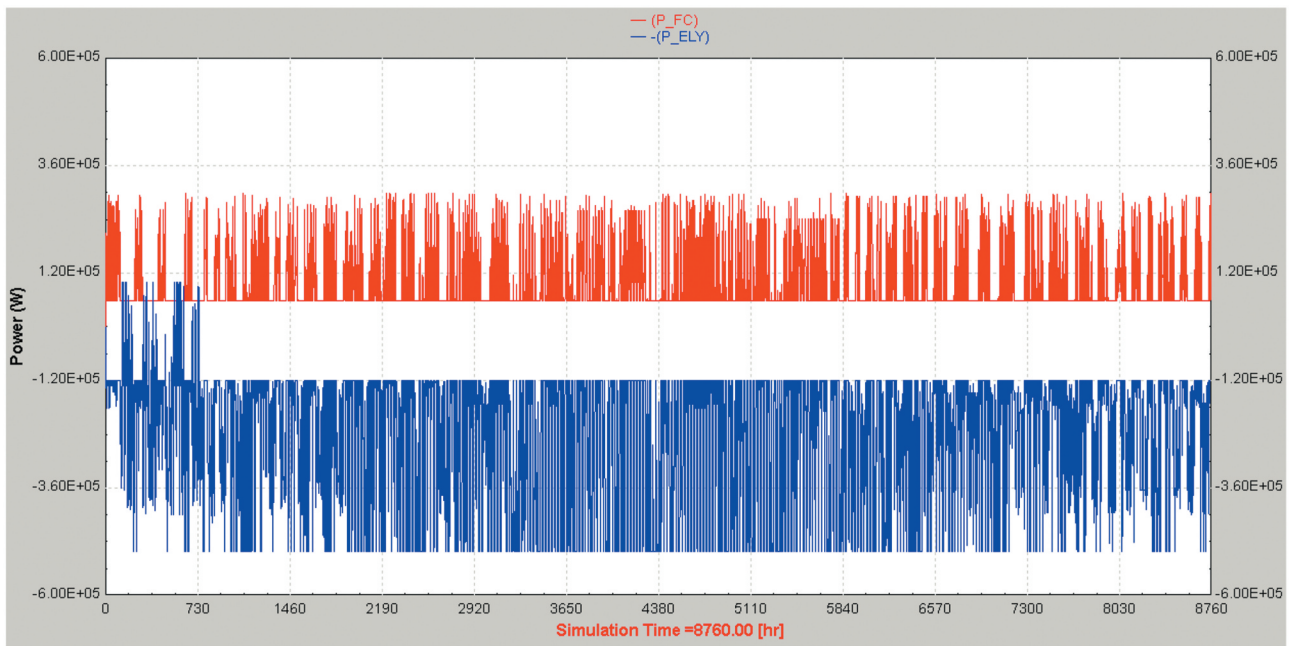


Figure 13. Hourly power output of the Alkaline Fuel cell (P_{FC}) and negative input power of the Electrolyzer ($-P_{ELY}$) relative to the optimal renewable hybrid system throughout the year.

is observed from this figure that, throughout the year hours, a lessening in dump power results in rise in the number of generator operating. A close scrutiny demonstrates that months with high renewable energy featured low commitment of generator, thus less number of operating diesel generators. Conversely, higher generator commitment occurred on months with lower renewable energy availability. For example, January month featured the lowest availability of renewable energy hence leading to higher

diesel generator contribution. This is expected because a diminishing in dump electricity signifies a reduction in the renewable energy sources responsible for power generation. This is observed especially in winter months (December-January-February) at which the number of required generators are between 3 and 5 at most time contrarily to hot seasons in which the penetration of renewable resources is higher given the solar radiation availability which could satisfy the load requirement. This

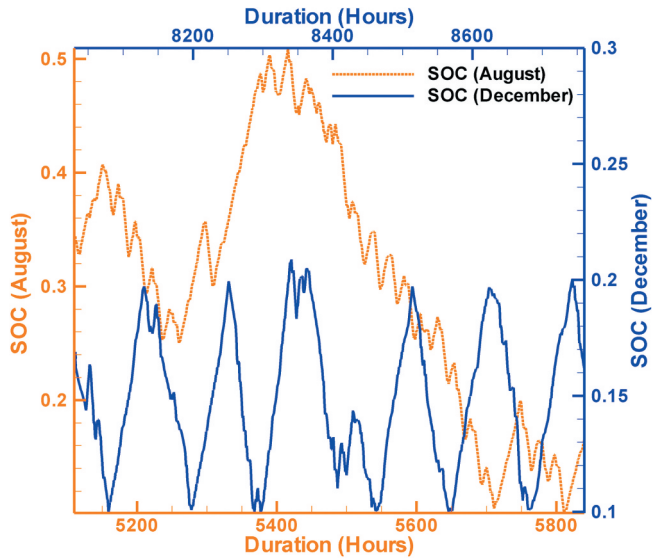


Figure 14. Temporal evolution of the Hydrogen Storage State of Charge (SOC) during two typical winter (December) and summer (August) months.

can be further explained as follows; in order to meet the load in times of reduced renewable energy sources (especially in cold period), the diesel generators are brought up (by the controller) to compensate for the decrease in renewable energy sources which in turn increases the required number of diesel needed for electricity production as backup .

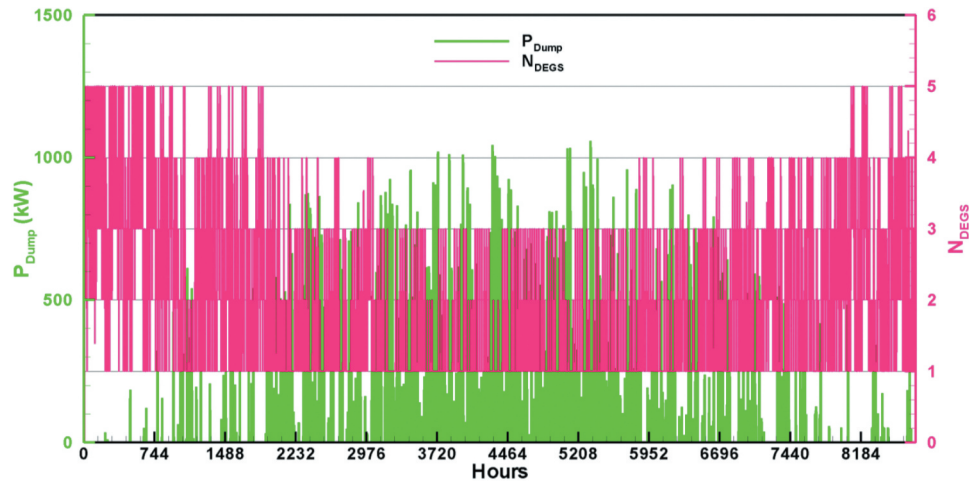


Figure 15. Illustration of the plot behavior of the dump power (P_{Dump}) and the required number of diesel generation (N_{DEGS}) during the year.

Table 2. Technical and economical details corresponding to the system components.

	Maximum output (kW)	Capital Cost (\$/kW)	Replacement (\$/kW)	Operation and maintenance costs (\$/kW)	Fuel cost (\$/m ³)	Life cycle (Year)	Capacity (kW)
PV panel	1167 kW	\$ 761.653/kW	\$ 761.653/kW	\$ 17.64/kW	-	20	1440 kW
Wind turbine	793 kW	\$ 1247.5/kW	\$ 1247.5/kW	\$ 26.656/kW	-	20	800 kW
Fuel Cell	295 kW	\$600/kW	\$600/kW	\$0.01/hour/kW = \$3.65/kW/year	-	10	300 kW
Hydrogen tank	916 (kg)	100 \$/m ³	100 \$/m ³	\$0.5/m ³	-	20	400 bar (60 m ³)
Electrolyzer	500 kW	\$150/kW	\$150/kW	\$8/year/kW	-	20	500 kW
Diesel Engines (5 single DEGS)	1500 kW	\$220/kW	\$200/kW	\$0.007/kWh	\$0.1/m ³	15000 h	300 *5 = 1500 kW
Converters	296 kW	\$550/kW	\$450/kW	\$5/kW/year	-	15	650 kW

5.2. Economical investigation

5.2.1. Technical and economical details

In the current section, a detailed economic examination is performed with the purpose to assess the financial and environmental profitability of the hybrid system under consideration.

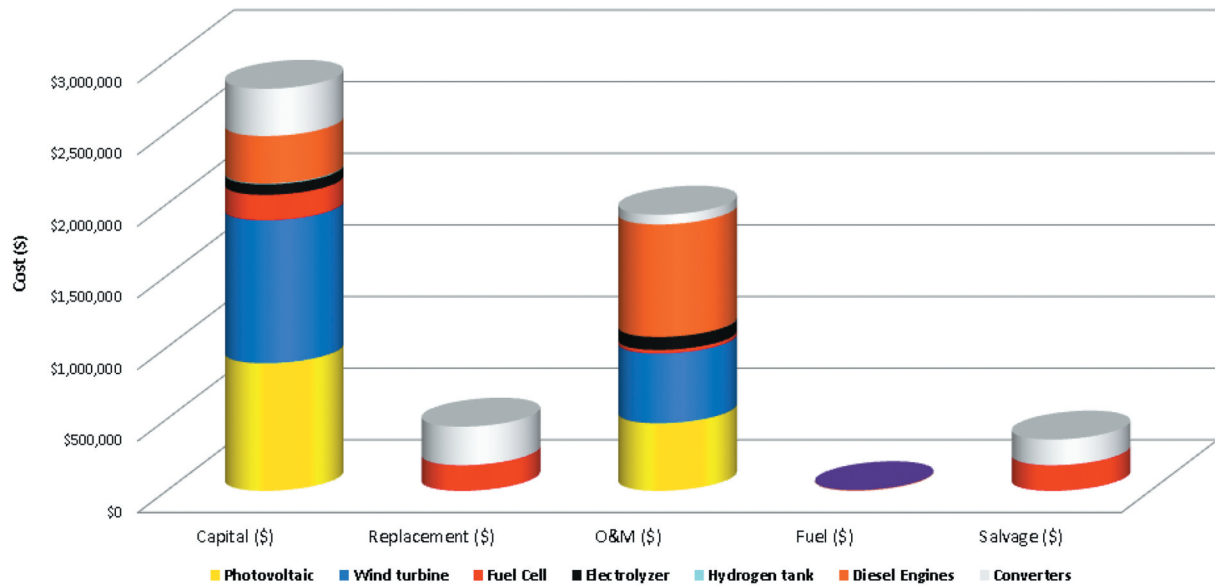
Table 2 summarizes the technical and economical details corresponding to the wind turbine, the PV array, the fuel cell system, the diesel generators, the converters, the electrolyzer and the hydrogen tank. Data were accurately and carefully collected to run the TRNSYS software for the 20 years of the project's life time.

The cost summary as well as energy outputs relatively to all components of the optimized hybrid renewable system are depicted in Table 3. The replacement cost of any of the energy sub-systems is evaluated during its lifetime by the simulations. The operation and maintenance cost of each component is considered as part of the economic study. Besides, the generator's operational cost comprises also the natural gas fuel cost throughout the lifetime of the generators.

The cost summary of the system by each component is shown in Figure 16 in terms of the capital cost, the cost of replacement, O&M, and the natural gas fuel as well as the salvage value. It is apparent that the operating and the replacement cost of the five single diesel engine generators set is higher than the capital cost contrarily to the remaining components in which the most investment is relative to the capital cost. Furthermore, compared to the other components, the wind turbine and the PV system are more costly but may be

Table 3. Cost summary and energy outputs relative to all components of the optimized HRES.

	Initial Investment Cost (\$)	Operations and Maintenance Costs (\$)	Annual Fuel Costs (\$)	Annual electricity output (kWh)	Replacement (\$)
PV panel	\$ 888,532	\$ 20,579	-	2,142,586 kWh	\$0
Wind turbine	\$ 998,003	\$ 21,325	-	2,304,950 kWh	\$0
Fuel Cell	\$176,962	\$ 1,077	-	852,722 kWh	\$176,962
Electrolyzer	\$ 75,000	\$ 4000	-	2,174,900 kWh	\$ 0
Hydrogen tank	\$ 6,000	\$30	-	-	\$0
Diesel Engines Natural gas (5 DEGS)	\$330,000	\$ 34,128	\$ 194	4,875,540 kWh (1,935.81 m ³ of natural gas fuel consumption)	\$0
Converters	\$ 330,000	\$3,000	-	128,000 kWh	\$ 270,000
HRES	\$ 2,804,497	\$ 84,139	\$ 194	10,200,000 kWh	\$ 446,962

**Figure 16.** Cost summary of the optimized HRES components.

projected to decline over time in the near future years due to the fact that renewable energy costs is surprisingly reducing year by year. This seems in accordance with cost type summarized in Figure 17(a) depicting the detailed cash flow simulated for the system. As observed, the wind turbines and PV array ranked the highest with a capital cost of 998,003 USD and 888,532, USD respectively. Nevertheless, as stated by International Renewable Energy Agency (IRENA 2016) (an organization bringing together 150 states), producing renewable energy costs less and less and prices have fallen by more than 10% in one year since 2009; up to 26% for the solar part as an example. As plotted in Figure 17(a), the positive sign of salvage value shows that there is a return after the end of the useful project lifetime calculated to be 356,962 USD. In the sub-Figure 17(b), we sketch the net present value (NPV) of total outputs with respect to each component during the 20 years life time of the system. As expected, an exponential decreasing trend is exhibited by the NPV outputs with respect to all components during the life time of the hybrid system.

5.2.2 Levelized Cost of Energy and carbon emission impact

In order to analyze economically the proposed hybrid system components and compare them during their useful life, the calculation of Levelized Cost of Energy (LCOE) has been adopted. The LCOE parameter defines the net present value

of the total cost (NPV_{Cost}) of power generated divided by the NPV of total electricity outputs ($NPV_{production}$) produced by the HRES over its lifetime. Besides, to get more accurate prevision of the LCOE of the hybrid system during its life time operation, the discount rate (r) is used to discount the energy cash-flows.

The formula of LCOE is given as follows:

$$LCOE = \frac{NPV_{Cost}}{NPV_{production}} = \frac{\sum_{t=1}^n \frac{(I_t + (OM)_t + F_t)}{(1+r)^t}}{\sum_{t=1}^n \frac{(E_{production})_t}{(1+r)^t}} \quad (29)$$

Where, (I_t) is the initial cost of investment expenditures, (OM)_{*t*} stands for the operating and maintenance expenses and (F_t) refers to the fuel expenditures applicable for the diesel engines in the t^{th} year. Here, the total output of the generated power asset contains also ($E_{production}$)_{*t*} which is the power energy production in the t^{th} year. Besides, two last important factors are to be considered in the equation, namely the discount rate of the project (r) fixed to be 0.093 and (n) referring to the life of the HRES system that has been taken to be 20 years. It is to note that the O&M expenses undergo an annual growth rate of 2% during the HRES system life time.

Computations lead to a NPV_{Cost} of 4,025,910 USD and $NPV_{production}$ of 81,822,193 kWh which leads to a LCOE of the HRES to be 0.0492 USD/kWh as illustrated in Table 4.

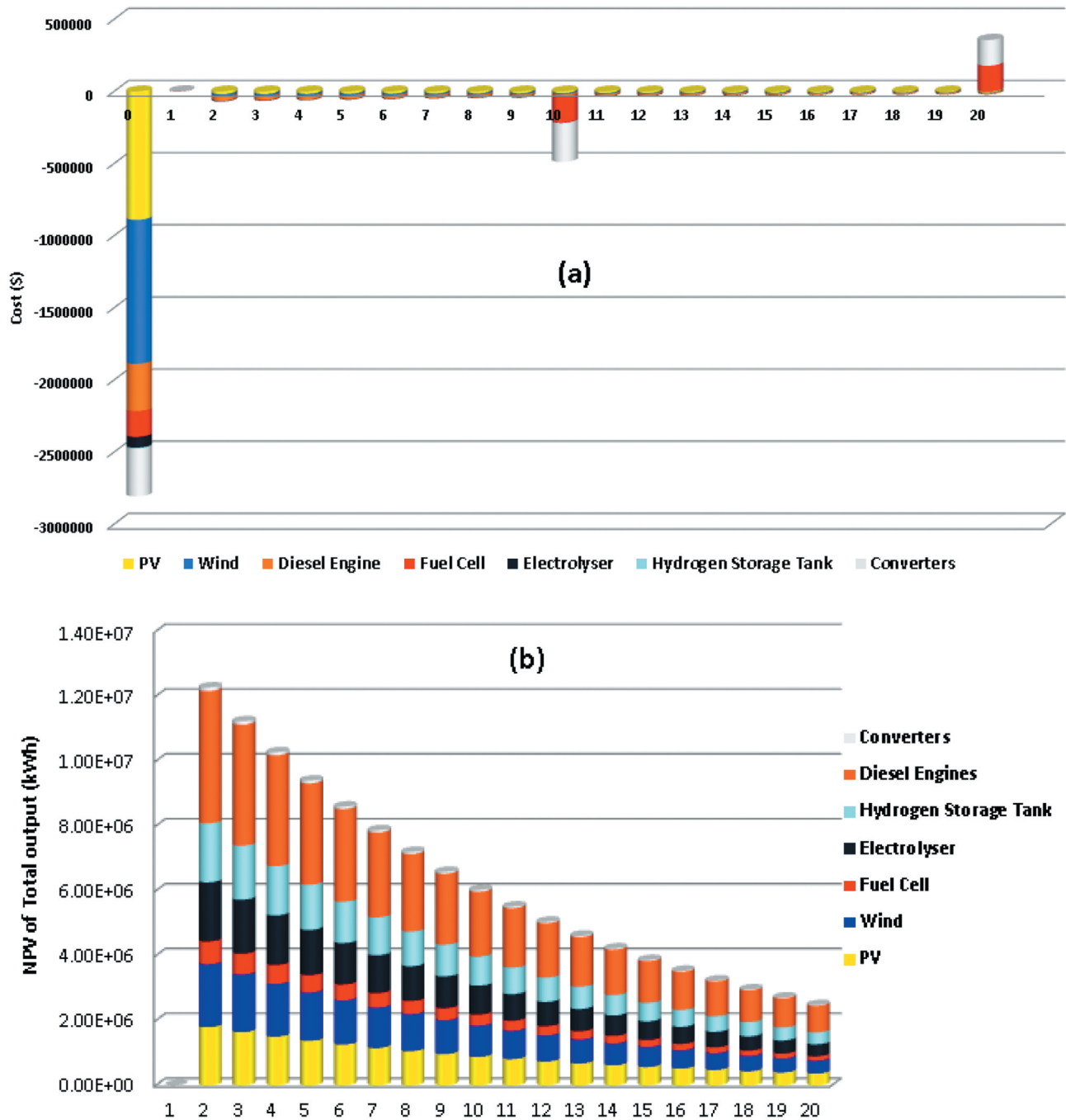


Figure 17. (a) Annual cash flow and (b) NPV of total outputs during the life time of the optimized system with respect to each component.

Table 4. Economic results of the optimized HRES system and emission reductions of CO₂.

NPV_{Cost} (\$)	$NPV_{production}$ (kWh)	LCOE (\$/kWh)	Emission Reductions Over 20 Years (t CO ₂)
\$4,025,910	81,822,193 kWh	\$0.0492/kWh	76,296 t CO ₂

Compared with other investigations in literature, the LCOE predicted by the proposed HRES in this study is less than that predicted by Zhao et al. (2018) who predicted a value of LCOE = 0.097\$/kWh in the year 2020 for

Tunisia with a photovoltaic and concentrating solar power system. If the decline in component prices continues as expected by IRENA (2016), standalone systems, as the HRES proposed herein, will be able to further compete with the grid energy in the near future.

On the other hand, the CO₂ emissions reduction per years life time is a techno-economic key index that also evaluates the performance of the HRES and its benefits on the environment. The carbon dioxide emissions are calculated by multiplying the quantity of total annual energy production during the life time of the HRES with the Emission Factor (EF) in Tunisia

estimated to be $EF = 0.374 \text{ tCO}_2/\text{MWh}$. Over 20 years of the system life time, an amount of 76,296 tons of CO_2 can be reduced with the renewable proposed system under study (Table 4).

6. Conclusions

In this paper, TRNSYS software is used to simulate a standalone hybrid renewable energy system consisting of wind turbine, a PV array, a fuel cell, an electrolyzer and diesel generators system as backup source. The hybrid micro grid power system is designed and optimized to meet the energy load demand relative to a residential community located in the village of Sidi Daoud situated in Tunisia. The hourly, daily and yearly performance and the components cost of the proposed off grid power system are predicted and analyzed. Among the goals of the utilized algorithm is to minimize the LCOE, diminish the dump and unmet electrical power and reduce the CO_2 emission as well. The numerical results demonstrate that the HRES meets the daily and yearly electrical demand of the household residential. For instance, typical weekly plots of HRES generation demonstrate that most of the electricity production during the daily hours is generated from the PV system, and the generator system seems to ensure the load demand during night with the fuel cell that only operates when needed. According to this analysis, for both winter and summer seasons, production peaks of the proposed system are attained at noon hours. Of great interest is the technical-economic analysis that is also carried out in the current study for cost summary of the system components in terms of capital cost, replacement cost, O&M, and fuel cost as well as the salvage value at the end of the life time of the HRES. Besides, annual cash flow, NPV of total outputs for all components, the LCOE, and the CO_2 reduction emission during the life time of the system is also analyzed and discussed. Finally, despite the fact that the adopted standalone hybrid system involves multiple renewable components with backup systems, computations lead to an acceptable renewable fraction (35.52%) and a low LCOE ($\$0.0492/\text{kWh}$) compared with that predicted in literature. The proposed HRES could be then more competitive knowing the continuous falling cost of renewable technologies, which may also lead to further rapid cost reductions of the proposed HRES in the near future.

Declaration of interest statement

The author declares that there are no conflicts of interest.

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